Hotel Booking Cancellation Prediction and Booking Segmentation

HE CAI [1] LI YUECHAO [1] LIANG SIMAN [1] NIU JINGQI [1] YOU YIRAN [1]

Abstract

We study hotel booking cancellation through a common workflow for data preparation, exploratory analysis, booking segmentation, classification, and model interpretation. The cleaned dataset contains 118,564 bookings from two Portuguese hotels, of which 37.26% were cancelled. Using the same 24 inputs and fixed training folds, random-holdout average precision is 0.373 for the constant-prior baseline, 0.801 for logistic regression, 0.911 for random forest, and 0.883 for XGBoost. The forest's F1 is 0.795 at threshold 0.50, compared with 0.803 at the training-selected threshold 0.45; it falls to 0.580 on unseen input profiles and to 0.539 in a separately tuned chronological check. Five descriptive segments show substantial variation in cancellation rates, while forest explanations highlight country, agent, lead time, and previous cancellations. The results support a retrospective decision-support benchmark, subject to field availability, repeated profiles, temporal transfer, calibration, and intervention costs.

1. Introduction

Hotel booking cancellations create uncertainty about how much reserved demand will become realised demand. A room released close to arrival may be difficult to resell, while staffing and other resources may already have been planned. A cancellation score can help prioritise booking confirmation and capacity review, but a useful analysis also needs to explain what the score measures, how it was evaluated, and which decisions the evidence can support.

This project combines the group's data preparation, exploratory analysis, K-means segmentation, baseline classification, ensemble modelling, explanation, and probability assessment into one analytical workflow. The aim is to compare predictive approaches under common inputs and evaluation definitions, then relate the resulting patterns to possible hotel decisions. Model comparison alone is insufficient: a high score on randomly held-out records can coexist with weaker performance on new booking profiles or a later arrival period.

We address four questions. **RQ1:** How well do the selected retrospective booking attributes discriminate cancellations under a shared holdout protocol? **RQ2:** Which attributes contribute to the fitted model's predictions? **RQ3:** How do cancellation patterns vary across hotel types and numerical booking segments? **RQ4:** How do calibration, threshold choice, and evaluation limitations affect the interpretation of operational recommendations? These questions preserve the proposal's predictive and explanatory aims while narrowing its initial-booking claim: the published data do not establish the timestamp at which every retained field became available.

Our contribution is an aligned comparison with traceable predictions and robustness checks. Segmentation describes booking combinations, classifiers predict cancellation, SHAP explains the fitted forest, and calibration assesses its scores. Neither a cluster label nor a feature attribution demonstrates an intervention effect.

2. Data and Preparation

2.1. Source and Analytical Population

The Hotel Booking Demand data describe bookings at a city hotel and a resort hotel in Portugal, with scheduled arrivals from July 2015 to August 2017 (António et al., 2019). The combined CSV contains 119,390 records and 32 columns. We retain the public source file alongside the cleaned file so that transformations can be traced. The binary target is `is_canceled`: 1 denotes cancellation and 0 denotes a booking that was not cancelled. The target is not a measured financial loss or a cancellation time.

The cleaning rules leave 118,564 records, a reduction of 826. The final file has no missing values and contains 44,176 cancellations and 74,388 non-cancellations. Table 1 gives the main sample counts. We preserve a unique `row_id` for matching data, folds, predictions, and diagnostic outputs. It is excluded from every predictive input matrix.

[1]Group 5, K6312 Information Mining and Analysis. Correspondence to: Group 5.

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Table 1. Cleaned data and fixed evaluation populations. Cancellation rates are calculated from unrounded counts.

Population	Bookings	Cancelled (%)
All cleaned data	118,564	37.26
City Hotel	78,899	41.91
Resort Hotel	39,665	28.01
Random training	94,851	37.26
Random test	23,713	37.26
Earlier-arrival training	96,428	36.48
Later-arrival test	22,136	40.64

2.2. Cleaning and Feature Construction

We fill missing children with zero and replace missing country and agent entries with an explicit unknown category. Agent is a categorical code, not an ordered numerical measurement. The highly incomplete company field is removed. Records with a zero guest count, a zero-night stay, or a negative average daily rate (ADR) are excluded under the predefined data-quality rules. The respective counts are 180, 715, and one, with overlapping conditions giving 826 excluded records. The rules are applied deterministically before the shared split; no quantile or scaling statistic is fitted on all records for supervised prediction. These exclusions define the study population and can also remove unusual legitimate bookings, a limitation of a rule-based cleaning policy.

We derive total guests as adults plus children plus babies, total nights as weekend nights plus weekday nights, a family indicator from the presence of children or babies, a lead-time band, and an arrival-season category. The last category is available for descriptive analysis but excluded from the common classifier. Derived fields share information with their components, so individual coefficients and attributions require care. The cleaned file retains 32 columns after removals and additions; this count includes the target, identifier, descriptive fields, and 24 supervised inputs.

We do not automatically delete identical-looking records. There is no true reservation identifier in the anonymised source, so matching records may represent separate reservations. The original raw file has 31,994 duplicate rows under its original columns, while the cleaned file has 33,985 duplicates when the artificial identifier is ignored. These are different definitions and should not be interpreted as the same count. Later analyses explicitly assess repeated 24-input profiles across evaluation partitions.

2.3. Common Predictors and Availability

The common classifier uses 15 numerical inputs and nine categorical inputs. Numerical inputs cover lead time, arrival-year/week/day, weekend and weekday nights, adults, children, babies, repeat-guest status, past cancellations, past non-cancelled bookings, total guests, total nights, and the family indicator. Categorical inputs are hotel, arrival month, meal, country, market segment, distribution channel, reserved room type, agent, and lead-time band.

We exclude direct outcome fields (reservation status and its date), fields that can change after booking (booking changes, assigned room type, and waiting-list days), company, and the common protocol's additional exclusions: ADR, customer type, deposit type, special requests, arrival season, and requested parking spaces. A field retained for EDA is therefore not automatically a classifier input. This distinction prevents an attractive descriptive association from silently changing the main model comparison.

Parking deserves explicit discussion. All 7,393 cleaned records requesting at least one parking space are non-cancellations; 7,360 request exactly one space. This near-deterministic pattern motivates precautionary exclusion, but does not itself establish when the field was recorded. The source construction refers to booking records reconstructed before arrival rather than guaranteed initial-booking snapshots (António et al., 2019). Removing parking and obvious outcome fields reduces risk without proving that every remaining attribute was available at the intended decision time.

2.4. Preprocessing and Partitioning

Numerical features are standardised and categorical features are one-hot encoded, with categories below 50 training occurrences pooled by the shared encoder. Unknown categories are handled by that encoder. We clone the unfitted preprocessing template within each model pipeline and fit it separately in each training fold. Standardisation is especially relevant to logistic regression and is retained for all classifiers to preserve the common representation. The random-training template produces 221 encoded features; that number may change with a different fitting population.

The primary split is stratified 80/20 with seed 42. Five fixed stratified folds partition only the 94,851 random-training records. We verify fold membership by `row_id`, rather than trusting positional alignment across exports. The supplementary scheduled-arrival split trains through 30 April 2017 and tests on May–August 2017. Its cutoff was selected from prespecified month-ends using sample balance and coverage, not predictive performance. Random-test labels are not used to fit preprocessing, tune classifiers, fit calibration maps, or choose thresholds.

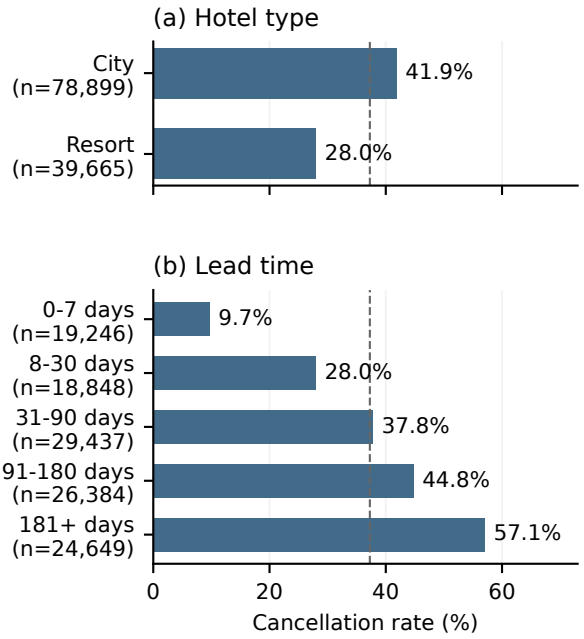


Figure 1. Descriptive cancellation patterns in the full cleaned data. Dashed lines mark the overall rate of 37.26%. Hotel and lead-time comparisons are associations rather than intervention effects.

3. Exploratory Analysis and Segmentation

3.1. Descriptive Cancellation Patterns

The full cleaned dataset has a cancellation rate of 37.26%; City Hotel's rate is 41.91% compared with 28.01% for Resort Hotel. Figure 1 also describes cancellation by lead-time band. This analysis describes the supplied population, including held-out records, and is not an independent validation experiment. The model feature list and fixed partitions are retained while exploratory associations are reported.

Deposit type illustrates why association must be distinguished from a recommendation. Non-refundable bookings have 14,494 cancellations among 14,587 records (99.36%), whereas bookings without a deposit have 29,646 cancellations among 103,815 records (28.56%). These conditional rates can reflect booking policies, group arrangements, coding conventions, or selection into contract types. They do not imply that imposing a non-refundable deposit causes cancellation. Deposit type is excluded from the shared classifier, and the analysis does not estimate the effect of changing a contract.

3.2. Training-Only K-Means

We cluster numerical booking attributes to obtain an interpretable segmentation separate from supervised prediction. The five inputs are lead time, total nights, total guests, ADR, and $\log(1+\text{previous cancellations})$. ADR is justified here as a segment descriptor even though it is outside the main classifier. Cancellation, country, and the row identifier do not form the clusters.

Table 2. Five-cluster profiles. Counts and rates summarise the full cleaned population after training-only fitting; means use original, uncapped units.

Cluster	n	Lead days	Nights	Cancel %
0	59,941	45.7	2.47	28.45
1	18,144	76.1	3.59	35.95
2	19,747	262.8	3.06	51.32
3	14,327	126.9	8.23	32.04
4	6,405	197.0	2.65	91.73

Outlier caps are learned from random training only: ADR 252, total guests 4, lead time 476 days, and total nights 14. The transformed inputs are standardised using training statistics. K-means is fitted to training records and then assigns the test records to the nearest learned centroid. Full-data profiles combine these assignments after fitting; they do not indicate that the model was fitted on the test set.

We compare alternative cluster counts. A three-cluster solution has silhouette 0.3165, compared with 0.3032 for the retained five-cluster solution. For five clusters, fits with alternative random seeds have adjusted Rand agreement of 0.9933–0.9960 relative to seed 42 in the training stability check. We retain five clusters for the more detailed, stable description; we do not call it the silhouette-optimal choice or a proven natural partition. Cluster numbers are arbitrary identifiers, and descriptive names are assigned from the fitted profiles, not their cancellation rates.

The largest cluster comprises shorter-lead, shorter-stay bookings. Cluster 1 has a mean party size of 2.91 and ADR of 175.83, compared with party sizes below two and ADR near 76–92 in most other clusters. Cluster 2 concentrates longer lead times, while cluster 3 separates long stays. Cluster 4 has mean prior cancellations of 1.61, alongside its much higher observed cancellation rate. These summaries do not mean that every member has the average characteristics.

Hotel composition changes some pooled comparisons. Long-stay bookings have a lower overall cancellation rate than the higher-ADR/larger-party segment (32.04% versus 35.95%), but within City Hotel this order reverses (41.62% versus 36.54%). Within Resort Hotel the corresponding rates are 27.40% and 35.00%. The long-stay segment is disproportionately Resort Hotel (67.3%), so an overall segment ranking should not be interpreted as a common ranking within both hotels.

The random-test segment rates are 28.63%, 36.18%,

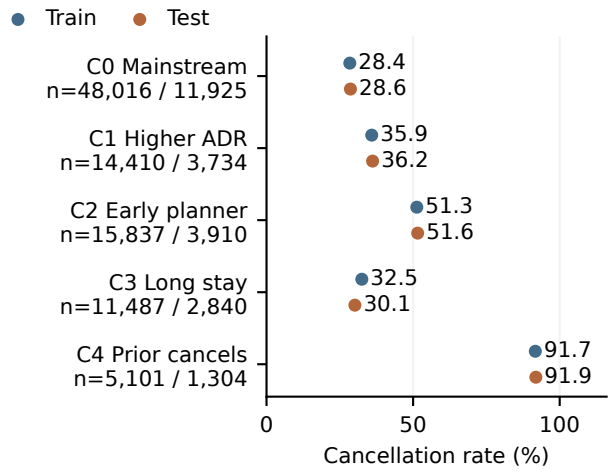


Figure 2. Cancellation rates after assigning records to training-fitted clusters; n gives training/test counts. Similar patterns describe a random split; they do not establish temporal transportability.

51.56%, 30.14%, and 91.87%, with corresponding counts 11,925, 3,734, 3,910, 2,840, and 1,304. Figure 2 compares training and test descriptions. Broad rate ordering persists in this random split, but this is not a test of future-period segment stability. Prior cancellation history was a clustering input, so the high cancellation rate in cluster 4 should not be presented as a wholly unexpected discovery.

4. Predictive Methods and Evaluation

4.1. Baselines and Candidate Models

The Dummy classifier estimates the training cancellation proportion as a constant probability. Since the proportion is below 0.50, all its decisions at threshold 0.50 are non-cancellations. Its accuracy is useful as a minimum decision baseline, while its average precision equals the evaluated cancellation prevalence when every score is tied.

Logistic regression models the log odds as a linear function of the encoded inputs with L2 regularisation. The common candidate grid varies $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and class weighting. We use the one-standard-error rule: among configurations within one standard error of the largest mean fold AP, choose the smallest C, representing stronger regularisation. The highest mean AP is 0.796712 at $C = 100$; its standard error is 0.001229, giving an eligibility cutoff of 0.795483. The selected configuration is $C = 1$ with no class weighting (mean AP 0.796565). Its coefficients, predictions, and metrics all refer to this retained estimator.

Random forests average decision-tree predictions and can represent nonlinear interactions (Breiman, 2001). We compare six prespecified combinations with 160 trees and seed 42, selecting by mean validation AP. The selected random-training configuration has depth 24, minimum leaf size 2, and maximum feature fraction 0.35. No class weighting is used. Its mean fold AP is 0.908, close to pooled out-of-fold AP, but those aggregations are not identical quantities.

Gradient-boosted trees sequentially add trees to improve a regularised objective (Chen & Guestrin, 2016). The XGBoost experiment uses the same data, inputs, training folds, and metric definitions. We compare three configurations: depth 3 with 200 trees, depth 5 with 300 trees, and depth 7 with 300 trees, all at learning rate 0.05. Common settings are histogram tree construction, row and column subsampling of 0.85, minimum child weight 1, L2 penalty 1, and seed 42. The depth-7 configuration has the highest mean fold AP, 0.882136. The grids differ between algorithms, so results compare these specified candidates rather than an exhaustive or compute-matched search.

4.2. Metrics and Selection Boundaries

We report precision, recall, F1, ROC-AUC, average precision (AP), Brier score, and confusion counts with cancellation as the positive class. Precision is $TP/(TP + FP)$, recall is $TP/(TP + FN)$, and F1 is their harmonic mean. The baseline decision threshold is fixed at 0.50 for every classifier. Changing the threshold affects decisions and confusion counts but not the underlying ranking metrics.

The column named `pr_auc` in the shared exports means average precision,

$$\mathrm{AP} = \sum_j (R_j - R_{j-1}) P_j, \tag{1}$$

using scikit-learn's non-interpolated convention (Scikit-learn Developers, 2026a). It is not trapezoidal PR-curve area. Brier score is $n^{-1} \sum_i (p_i - y_i)^2$, where lower is better. It reflects both calibration and discrimination, so a lower Brier score alone does not fully diagnose calibration. ROC-AUC assesses ranking across both classes; AP also depends on class prevalence.

Candidate selection and out-of-fold prediction reuse the common training folds. The selected-model validation numbers are therefore post-selection summaries, not nested-CV estimates. Similar validation and test metrics are reassuring descriptions, but do not prove the absence of overfitting, especially with repeated profiles. Alternative thresholds and calibration maps are fitted using training-derived predictions only and evaluated separately from the baseline table.

Table 3. Aligned test results at threshold 0.50. AP means average precision. Shared later-arrival rows reuse randomly selected settings and are not an untouched future holdout. The separately selected forest has a different selection protocol.

Evaluation	Model	Precision	Recall	F1	ROC-AUC	AP	Brier
Random test	Dummy	0.000	0.000	0.000	0.500	0.373	0.234
	Logistic regression	0.746	0.651	0.695	0.867	0.801	0.142
	Random forest	0.843	0.752	0.795	0.937	0.911	0.097
	XGBoost	0.804	0.715	0.757	0.919	0.883	0.111
Shared later-arrival	Dummy	0.000	0.000	0.000	0.500	0.406	0.243
	Logistic regression	0.634	0.740	0.683	0.802	0.706	0.181
	Random forest	0.726	0.553	0.628	0.836	0.762	0.162
	XGBoost	0.707	0.652	0.679	0.847	0.777	0.155
Separate time selection	Random forest	0.810	0.404	0.539	0.828	0.764	0.172

5. Aligned Model Results

5.1. Random Holdout

The random forest improves materially over logistic regression in the retained random holdout. Its F1 is 0.795, precision 0.843, recall 0.752, ROC-AUC 0.937, and AP 0.911. Logistic regression obtains F1 0.695 and AP 0.801; the constant-prior baseline has F1 zero and AP 0.373. The forest's 2,191 false negatives and 1,240 false positives mean that 24.8% of cancelled bookings are missed at 0.50, despite an overall accuracy of 85.5%. This is why accuracy alone is insufficient.

XGBoost obtains random-test F1 0.757, ROC-AUC 0.919, and AP 0.883. The random forest has the highest mean training-fold AP among the tested model configurations, so we retain it as the principal benchmark and explanation model. The held-out results are reported after that selection rather than used to choose an algorithm. This choice does not imply that more extensive boosting optimisation would also favour the forest.

The no-agent sensitivity changes random-test AP from 0.801 to 0.784 for logistic regression and from 0.911 to 0.898 for the forest; logistic F1 changes from 0.695 to 0.686. The ablations retain each main model's settings and are not independently retuned. They support retaining agent in this benchmark subject to availability and transfer checks, without showing that an agent causes cancellation. Agent and country are contextual codes whose associations may depend on hotel and period. Their predictive value should be reassessed when the booking mix changes.

5.2. Later-Arrival Evaluation

Under the shared team's later-arrival comparison, logistic regression has F1 0.683 and AP 0.706, whereas the forest has F1 0.628 and AP 0.762. Better ranking and better decisions at one fixed threshold can therefore select different models. The forest's recall is only 0.553 on this period, compared with 0.740 for logistic regression. A statement that the forest is best on every metric would be incorrect.

XGBoost's corresponding later-arrival F1 is 0.679 and AP is 0.777, with Brier score 0.155. This common time comparison has an important selection limitation: 17,707 of the 22,136 later-arrival test rows also belong to the random-training set used to choose model configurations. Their outcomes have therefore informed parameter selection. Refitting on earlier dates alone does not erase that reuse. We retain these scores as a clearly labelled comparison under the team's original procedure and do not describe them as an untouched future holdout or use them to revise the principal model choice.

A separate forest experiment selects parameters only through five expanding windows of unique scheduled arrival dates within the earlier-arrival training period. Every validation date follows its corresponding fitting dates. The independently selected forest uses depth 16, minimum leaf size 5, and `sqrt` features, then refits on all early records. On the later-arrival test it obtains F1 0.539, precision 0.810, recall 0.404, ROC-AUC 0.828, and AP 0.764. This is a more defensible selection boundary for this forest, although scheduled arrival order is not a complete reconstruction of booking-time information availability. The difference between the two time scores also includes a change in selected parameters and cannot isolate the effect of outcome reuse.

5.3. Repeated-Profile Sensitivity

In the random holdout, 8,998 of 23,713 test records (37.9%) have an identical 24-input profile in training. Forest F1 is 0.959 on these profiles but 0.580 on the 14,715 unseen profiles; corresponding AP values are 0.992 and 0.685. Equal weighting of unseen profiles gives F1 0.579. In addition, the common folds contain 6,660 input profiles that cross fold boundaries, involving 34,143 training rows. Random

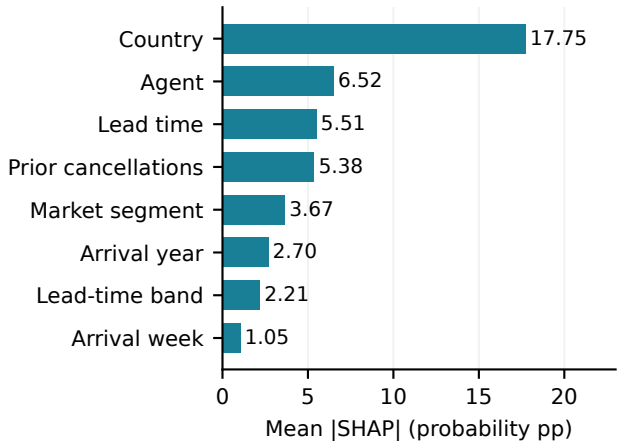


Figure 3. Grouped global SHAP importance for the fixed random-training forest. Units are probability percentage points. These explanations are specific to this estimator and sample.

validation therefore partly evaluates familiar patterns.

These diagnostics do not establish that all repeated profiles are erroneous records. They change the evaluation population and target prevalence, so the score gap is not a causal estimate of duplication's effect. They do show why an unqualified random-holdout score is too broad a deployment claim. A fully group-disjoint tuning protocol and prospective evaluation would be appropriate next steps. We keep the shared split intact to preserve comparability rather than deleting difficult records after observing performance.

6. Interpretation of the Random Forest

6.1. Global Contributions

We explain the saved random-training forest using exact interventional Tree SHAP for cancellation-class probability, following the additive attribution framework of Lundberg & Lee (2017). The background is 100 random-training bookings sampled with seed 42, and the explanation sample is 400 random-test bookings sampled with seed 2026. We sum one-hot attributions within each original input before averaging their absolute values. Thus, a categorical variable's importance is measured on the original-variable scale rather than inflated by counting its encoded columns separately.

The five largest mean absolute contributions are country (17.75 percentage points), agent (6.52), lead time (5.51), previous cancellations (5.38), and market segment (3.67). These explain the fitted model's score variation relative to its background; they do not estimate causal effects or the accuracy gained by adding each variable. Lead time and its band, calendar fields, stay components, and guest-composition features share information, which can divide attribution among correlated inputs. Interventional perturbations may also create combinations that are rare in actual bookings.

The numerical SHAP representation aligns tree thresholds with the forest's float32 routing without altering fitted trees or scores. Reconstructing probability as the background plus all feature contributions has maximum absolute error 4.49×10^{-8} across the 400 explained records. This confirms numerical reconstruction only. It does not validate calibration or prove that the explanation describes a feasible intervention.

6.2. Individual Examples and Scope

Two examples are chosen as the lowest and highest forest scores in the explanation sample, without selecting on outcome. Booking 3423 has a raw score near zero and outcome 0. Against a background of 37.38%, its four largest absolute contributions are country ESP (-20.69 percentage points), arrival year 2015 (-6.14), no previous cancellations (-4.10), and agent 38 (-3.77). Booking 75648 has a raw score near one and outcome 1; its largest terms are one previous cancellation ($+34.90$), country PRT ($+20.61$), lead time 290 days ($+5.23$), and arrival year 2015 (-4.27). The remaining terms are still needed for exact reconstruction.

These examples illustrate a decomposition, not typical predictive accuracy or certainty about an individual. Country and agent values encode context and may proxy policies, demand composition, or other unmeasured factors. They should not be treated as inherent guest characteristics or grounds for punitive decisions. The figures explain the random forest even if another classifier ranks better; they must not be relabelled as explanations of a boosted-tree model.

7. Probability Assessment and Thresholds

7.1. Training-Derived Probability Assessment

Ranking, calibrated probability, and a binary decision are separate outputs. We assess the raw probabilities of logistic regression, the random forest, and XGBoost on their common random-test population. Reliability diagrams compare average scores with observed cancellation proportions within ten equal-frequency bins; repeated scores can reduce the number of distinct bins. The diagonal is a calibration reference, while departures describe the fixed model on this particular population (Scikit-learn Developers, 2026b).

We also evaluate a sigmoid mapping, fitted by univariate logistic regression to the clipped logit of training out-of-fold scores (probabilities clipped to $[10^{-6}, 1-10^{-6}]$, $C = 10^6$). For the validation threshold analysis, this mapping is cross-fitted over the existing five OOF folds: each validation fold

Table 4. Random-test decisions at training-selected raw-score thresholds. Thresholds optimise OOF F1, not revenue.

Model	Threshold	Precision	Recall	F1
Logistic	0.35	0.657	0.814	0.727
Forest	0.45	0.795	0.811	0.803
XGBoost	0.40	0.720	0.838	0.775

is mapped using a sigmoid fitted to the other folds' OOF score–label pairs. A final mapping fitted to all training OOF pairs transforms random-test scores. No random-test labels fit the mapping or determine a threshold.

This is a practical post-selection assessment, not fully nested calibration validation. Base OOF models share fitting records, candidate selection has already used these folds, and the OOF score distribution may differ from scores of the model refitted on all training data. Cross-fitting the sigmoid reduces direct reuse of a row's label by its own mapping, but does not remove these dependencies. We do not select a new model using the calibration test results or claim that the mapped probabilities are validated for a future period.

At threshold 0.50, sigmoid mapping changes Brier scores from 0.141971 to 0.141964 for logistic regression, 0.097321 to 0.096971 for the forest, and 0.111488 to 0.110991 for XGBoost. Improvements are small; the strongest raw benchmark already has substantially lower Brier error than the linear baseline. Since each fitted mapping is monotone increasing, ROC-AUC and AP remain unchanged in these test outputs. Calibration changes score interpretation and potentially decisions, not the order of bookings in this experiment. Brier scores are independent of the chosen decision threshold.

7.2. Validation-Selected Thresholds

For raw and sigmoid scores separately, we search thresholds 0.05–0.95 in increments of 0.05, maximise training OOF F1, and break ties in favour of the smaller threshold. This selects raw thresholds 0.35 for logistic regression, 0.45 for the forest, and 0.40 for XGBoost. We evaluate these already fixed rules on the random test. We do not report a test-optimised threshold or replace the 0.50 baseline results with tuned results in the main comparison.

For the forest, changing the threshold from 0.50 to 0.45 raises recall from 0.752 to 0.811 and F1 from 0.795 to 0.803, while precision falls from 0.843 to 0.795. False negatives decrease from 2,191 to 1,666, but false positives increase from 1,240 to 1,854. This creates 525 additional correctly flagged cancellations alongside 614 additional false alarms. Whether that trade-off is worthwhile depends on the review action, staff capacity, and guest contact costs.

The sigmoid models select the same coarse-grid thresholds as their raw counterparts, with test F1 values 0.727, 0.803, and 0.775 respectively after rounding. Their small differences do not justify a claim of a large classification benefit. For the benchmark, we retain the raw forest and report 0.50 as the common comparison threshold, with 0.45 as a separate candidate review rule. Neither threshold is established as optimal for a hotel's economic objective or for the later-arrival population.

8. Discussion and Operational Implications

The research questions lead to a qualified but coherent conclusion. For RQ1, the common retrospective attributes support useful cancellation ranking within the random benchmark, and nonlinear models improve on the linear baseline in that population. However, repeated profiles and the independently tuned time check reveal a narrower scope than the random score suggests. For RQ2, forest scores are strongly associated with booking context, lead time, and historical cancellations. For RQ3, hotel type and the five numerical segments show distinct observed cancellation rates. For RQ4, a ranking score needs calibration assessment and an explicit decision policy before it can become an operational trigger.

The most immediate use case is prioritising a small set of bookings for confirmation or staff review. Longer-lead segments can motivate earlier reminders, while long-stay segments can inform capacity review. These are candidate actions to test, not measured benefits from this study. We have neither intervention outcomes nor reliable booking-level cancellation costs, so we cannot quantify recovered revenue, recommend an optimal deposit amount, or claim that a policy reduces cancellations.

With calibrated probabilities and known error costs, a simplified review rule would flag a booking when $p > C_{FP}/(C_{FP} + C_{FN})$, assuming correct decisions have zero cost and the stated false-positive and false-negative costs apply. Actual review, contact, room replacement, and guest experience costs complicate that rule. An F1-selected threshold is a reproducible statistical benchmark rather than a revenue-maximising policy. Calibration on historical random-training predictions also does not guarantee calibration after temporal drift (Scikit-learn Developers, 2026b).

Several limitations remain. The observations come from two hotels in 2015–2017, so transfer to another hotel or period requires new evaluation. The source is retrospective and lacks a complete initial-booking snapshot, which prevents a verified real-time prediction claim. Full-data EDA is descriptive, and shared folds are not group-disjoint. Hyperparameter and threshold summaries are post-selection,

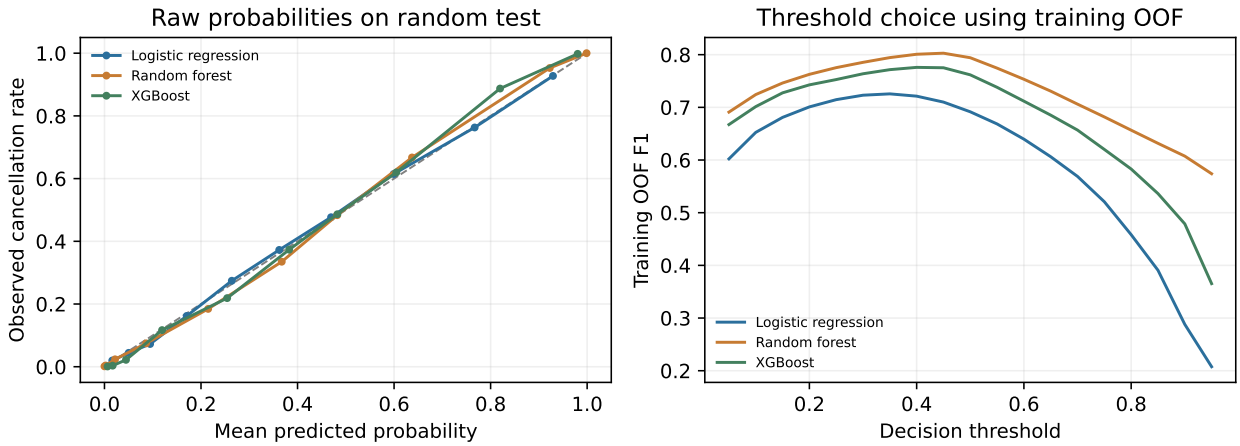


Figure 4. Left: raw-score reliability on random test after all models are fixed. Right: threshold curves from training OOF scores only. Test labels are used to evaluate calibration, not to choose a threshold.

not nested validation. The independently tuned time experiment covers the forest only and does not establish a clean chronological ranking of all four algorithms. Changes in prevalence also affect AP and calibration.

Finally, the data do not identify causal drivers of cancellation or the effect of changing guest attributes. Country and agent may encode sensitive contextual patterns; automated adverse treatment would be unsupported. A future operational study should establish feature timestamps, use group-aware time-respecting selection, validate on newer records, measure review capacity and error costs, and evaluate reminders prospectively. Transparent checks on evaluation boundaries are necessary because leakage can otherwise produce misleading performance claims (Kapoor & Narayanan, 2023).

9. Conclusion

We provide a common analytical record linking data cleaning, descriptive booking patterns, training-fitted segmentation, classifier comparison, forest explanation, and probability assessment. Cancellation rates differ substantially across hotels and booking segments. The random forest outperforms the logistic baseline in the retained random benchmark, while unseen-profile and separately tuned time results caution against extending that finding to unfamiliar bookings. Model ranking, a thresholded decision, and an explanation answer different questions. The defensible practical conclusion is to treat these models as retrospective candidates for locally validated decision support, with field availability, prospective performance, and intervention value still to be established.

Code and Data Availability

Code and data are distributed in the accompanying reproducibility archive.The accompanying project includes the source data, fixed row identifiers and validation folds, executable notebooks and scripts, prediction files, model settings, and the tables used in this report. The underlying public dataset is attributed to António et al. (2019). Individual booking identifiers are unavailable; the project-specific `row_id` supports reproducible alignment only.

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A. Reproducibility Checks

The analytical record separates reference experiments from integration checks. Data cleaning was rerun from the raw CSV, reproducing all 32 cleaned columns with zero differences. Random train/test identifiers were reproduced in their supplied order, and all 94,851 fixed validation-fold assignments matched. The selected five-cluster fit was rerun, reproducing its original cluster sizes; the alternative-k and seed diagnostics remain the executed notebook record.

For Dummy, logistic regression, its no-agent variant, random forest, and its no-agent variant, all saved prediction rows were checked against the common data and split identifiers. Recomputed metrics matched to at most 1.4×10^{-16}. Re-prediction with the saved random-training estimators differed from six-decimal exports by at most 5.01×10^{-7}, consistent with rounding. The saved chronological forests also matched full-precision exports, and all five independent chronological validation windows preceded the test period. The SHAP reconstruction check covers all 400 explained rows.

The completed XGBoost study fitted all three configurations within the fixed five folds, then fitted its selected random and earlier-arrival models. Independent checks reproduced its metrics, configuration selection, 114 raw/sigmoid validation-grid rows across three models, and 12 test-assessment rows. The sigmoid maps were refitted from the training OOF exports and their test transformations checked. These checks do not imply that all historical model hyperparameter searches were fully retrained during integration or that the OOF protocol is nested.

Source-built preprocessing reproduces the original template's 221 encoded feature names and all random-test transformation values exactly. This avoids requiring the historical template pickle for new fitting. The integration environment used Python 3.11.2 and scikit-learn 1.9.1; the historical forest environment records scikit-learn 1.9.0, while the logistic regression metadata record 1.9.1. Original environment metadata are retained. Compatibility was verified for the recorded loading environment rather than assumed for arbitrary package versions.

The principal executable checks are `src/verification.py` and `src/verify_e.py`. New training through `src/run_project.py` writes to a separate run directory. The standard prediction key is model, variant, experiment, subset, and row identifier; validation records additionally carry the fixed fold number. Full-precision E probabilities and the six-decimal historical exports are both evaluated at their documented precision. This preserves an auditable relationship between the report, saved predictions, and code.